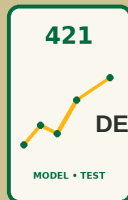
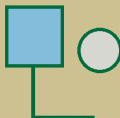
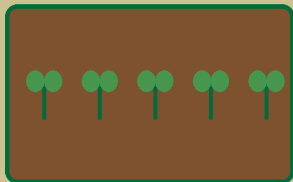


Frame a Causal Question



MODEL • TEST

FRAME A CAUSAL QUESTION

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

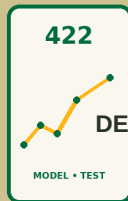
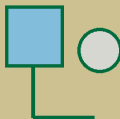
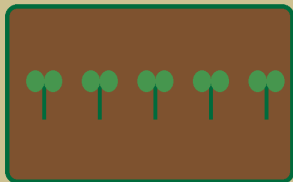
Complete the investigation using explicit assumptions, systematic measurements, and reproducible records. Interpret the evidence and state the decision implication.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE

Design a Factorial Experiment



DESIGN A FACTORIAL EXPERIMENT

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

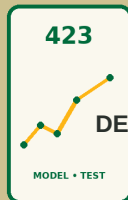
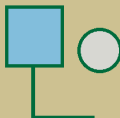
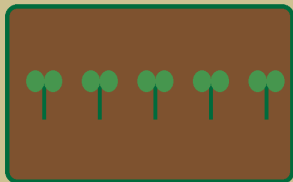
Define the objective, boundary, variables, constraints, and performance criteria. Develop a testable design or model and document the assumptions.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE

Identify Main Effects



IDENTIFY MAIN EFFECTS

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

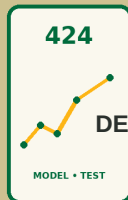
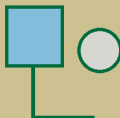
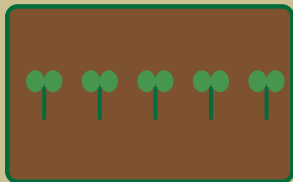
Complete the investigation using explicit assumptions, systematic measurements, and reproducible records. Interpret the evidence and state the decision implication.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE

Interpret an Interaction



INTERPRET AN INTERACTION

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

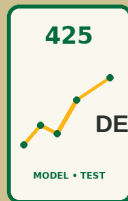
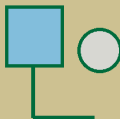
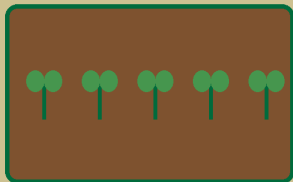
Use quantitative evidence and stated criteria. Compare alternatives or conditions, account for uncertainty, and justify the conclusion.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE

Detect Confounding



DETECT CONFOUNDING

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

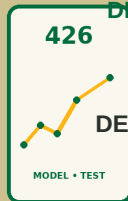
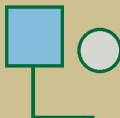
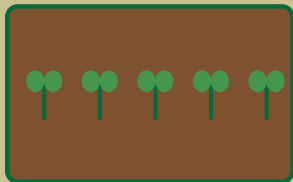
Identify the abnormal signal or performance pattern, formulate plausible causes, test them against independent evidence, and recommend a corrective action.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE

Develop a Blinded Measurement Plan



DEVELOP A BLINDED MEASUREMENT PLAN

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

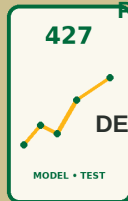
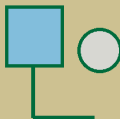
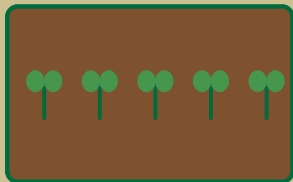
Define the objective, boundary, variables, constraints, and performance criteria. Develop a testable design or model and document the assumptions.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE

Perform an Instrument Calibration



PERFORM AN INSTRUMENT CALIBRATION

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

PA-427 • Perform an Instrument Calibration

TECHNICAL INVESTIGATION

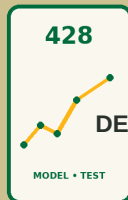
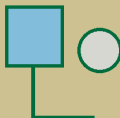
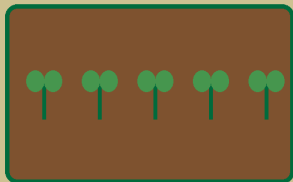
Use a documented protocol and appropriate instrument or reference.
Repeat where needed, record metadata and units, and assess data quality.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE

Quantify Repeatability



QUANTIFY REPEATABILITY

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

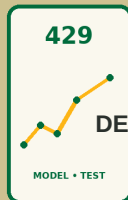
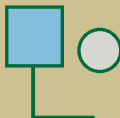
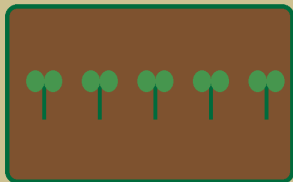
Use the specified measurements, equation, and assumptions. Show units throughout, calculate the result, and interpret its engineering or production meaning.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE

Conduct a Sensitivity Analysis



CONDUCT A SENSITIVITY ANALYSIS

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

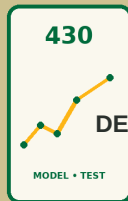
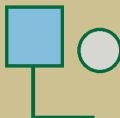
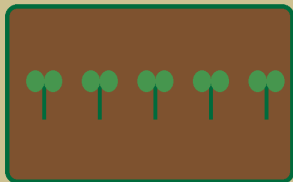
Complete the investigation using explicit assumptions, systematic measurements, and reproducible records. Interpret the evidence and state the decision implication.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE

Evaluate Causal Evidence



EVALUATE CAUSAL EVIDENCE

DESIGN • INSTRUMENT • MODEL • EVALUATE

DESIGN → INSTRUMENT → MODEL → EVALUATE → OPTIMIZE

TECHNICAL INVESTIGATION

Use quantitative evidence and stated criteria. Compare alternatives or conditions, account for uncertainty, and justify the conclusion.

EVALUATE & DECIDE

State the result, uncertainty or limitation, and the next design, measurement, or management decision supported by the evidence.

DEFINE → MEASURE → MODEL → TEST → DECIDE → IMPROVE